

SVICHTOV, Bulgaria

WMO: 155330

Lat: 43.617N

Lon: 25.350E

Elev: 26

StdP: 101.01

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.6	-12.8	1.3	-10.7	-10.3	1.6	-8.3	16.8	2.8	15.2	3.3	1.8	50	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	34.6	23.7	32.8	23.1	31.2	22.2	24.7	32.8	23.7	31.4	22.9	30.0	2.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	16.8	30.0	21.1	15.8	28.8	20.3	15.0	27.6	75.2	32.7	71.2	31.4	67.9	30.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	12.0	9.7	DB	-13.7	37.8	3.6	2.5	-16.2	39.6	-18.3	41.1	-20.3	42.5	-22.9	44.3
			WB	-14.0	26.8	3.5	1.2	-16.5	27.6	-18.6	28.3	-20.6	28.9	-23.1	29.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	0.0	2.5	7.9	13.5	18.9	22.9	25.2	24.9	19.6	13.1	6.8	1.0	(6)
(7)		DBStd	9.78	5.04	5.42	4.74	3.61	3.33	3.17	2.87	2.87	3.38	4.08	4.14	4.83	(7)
(8)		HDD10.0	1041	311	211	97	12	0	0	0	0	17	112	280		(8)
(9)		HDD18.3	2596	568	442	323	149	34	4	0	1	26	167	345	537	(9)
(10)		CDD10.0	2168	1	3	33	116	278	387	470	461	287	114	18	1	(10)
(11)		CDD18.3	680	0	0	1	3	53	141	212	203	63	5	0	0	(11)
(12)		CDH23.3	5916	0	0	2	38	385	1196	2016	1849	403	26	1	0	(12)
(13)		CDH26.7	2281	0	0	0	8	88	437	867	777	103	2	0	0	(13)
(14)	Wind	WSAvg	2.6	2.5	3.0	3.4	2.9	2.7	2.6	2.6	2.1	2.3	2.1	2.3	2.5	(14)
(15)	Precipitation	PrecAvg	564	44	33	39	43	62	61	51	41	57	49	36	48	(15)
(16)		PrecMax	907	119	99	79	85	136	170	167	130	166	124	100	157	(16)
(17)		PrecMin	334	3	3	0	3	16	6	7	4	5	0	0	6	(17)
(18)		PrecStd	137	25	21	20	21	29	37	40	32	41	33	24	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	17.0	22.7	27.6	31.0	34.8	37.6	36.9	31.9	26.2	20.1	15.0	(19)
MCWB			8.9	10.6	13.8	18.2	20.6	23.8	24.8	24.2	21.4	19.7	14.9	11.1	(20)	
2%		DB	10.9	14.2	20.1	23.6	28.9	32.6	34.9	34.4	29.4	23.4	17.0	12.1	(21)	
		MCWB	7.8	9.3	12.4	15.8	19.8	23.2	23.8	23.6	20.6	17.9	13.4	9.2	(22)	
5%		DB	8.7	12.1	17.8	21.4	27.3	31.0	32.9	32.5	27.5	21.0	14.7	9.0	(23)	
		MCWB	6.3	8.4	11.3	14.6	19.2	22.2	23.1	22.9	19.7	16.4	12.2	7.3	(24)	
10%		DB	6.6	9.9	15.6	19.6	25.5	29.3	31.1	30.9	25.6	19.0	12.7	6.9	(25)	
		MCWB	4.8	6.9	10.1	13.5	18.4	21.3	22.3	22.1	18.8	15.4	10.5	5.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	11.6	14.9	18.8	22.1	25.2	25.8	26.3	22.4	20.0	15.5	11.4	(27)
MCD B			12.7	15.7	20.9	25.6	28.6	32.9	35.0	34.8	30.3	25.3	19.0	14.7	(28)	
2%		WB	8.0	9.8	13.1	16.5	20.8	23.9	24.6	24.7	21.1	18.4	13.8	9.3	(29)	
		MCD B	10.5	13.6	18.8	22.4	27.4	31.3	33.4	32.4	28.1	22.6	16.4	11.8	(30)	
5%		WB	6.4	8.4	11.8	15.2	19.7	22.9	23.7	23.7	20.2	17.0	12.5	7.5	(31)	
		MCD B	8.6	11.7	17.2	20.4	26.0	29.8	31.6	31.0	26.4	20.3	14.4	8.9	(32)	
10%		WB	4.9	7.0	10.4	14.1	18.7	21.9	22.8	22.7	19.3	15.6	11.0	5.7	(33)	
		MCD B	6.5	9.8	15.2	18.6	24.4	28.3	30.0	29.7	24.8	18.6	12.6	6.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	6.5	8.7	9.6	10.4	10.3	10.5	10.6	9.3	7.7	5.4	4.7	(35)
MCD BR			7.7	10.5	12.7	12.5	13.0	12.6	13.0	12.9	12.4	10.6	8.5	8.4	(36)	
MCWBR			5.5	6.9	7.0	6.4	6.0	5.4	5.3	5.3	5.7	6.1	5.7	6.4	(37)	
5% WB		MCD BR	7.5	10.1	12.2	11.6	12.1	12.2	12.4	12.3	11.6	9.9	7.6	7.9	(38)	
		MCWBR	5.5	6.9	7.2	6.4	6.0	5.7	5.4	5.6	5.7	6.1	5.3	6.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.380	0.419	0.471	0.446	0.454	0.464	0.477	0.443	0.405	0.374	0.352	(40)	
taud		2.281	2.210	2.150	2.019	2.167	2.183	2.162	2.122	2.183	2.270	2.313	2.292	(41)		
Ebn at Noon		755	793	810	795	830	823	809	782	777	756	720	710	(42)		
Edh at Noon		87	111	134	165	146	144	146	147	128	104	85	79	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.48	2.29	3.42	4.75	5.88	6.44	6.52	5.83	4.25	2.75	1.57	1.20	(44)	
(45)		RadStd	0.22	0.29	0.33	0.49	0.47	0.44	0.43	0.35	0.33	0.31	0.22	0.17	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.61	+1.38	+1.11	N/A	N/A	N/A	N/A	(46)
(47)	Regional (48 neighbors)	+0.45	N/A	N/A	N/A	+0.42	N/A	N/A	-140	+86	N/A	(47)

Nomenclature: See separate page